

Mathematics

Very short answer type question

1) →

2) → The distance of a point (a, b) from origin = $\sqrt{a^2 + b^2}$

3) → Given,

$$\angle BAC = \angle ADC$$

In $\triangle ABC$ and $\triangle DAC$,

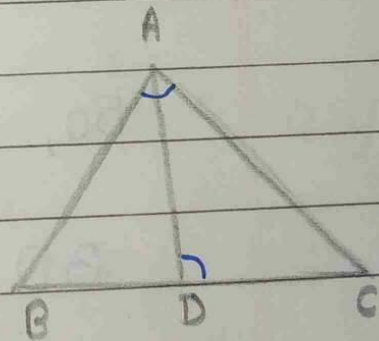
$$\angle BAC = \angle ADC - \text{Given}$$

$$\angle ACB = \angle ACD - \text{Common}$$

By AA similarity, $\triangle ABC \sim \triangle DAC$

$$\therefore \frac{AC}{BC} = \frac{DC}{AC}$$

$$\therefore \frac{CA}{CB} = \frac{DC}{AC}$$

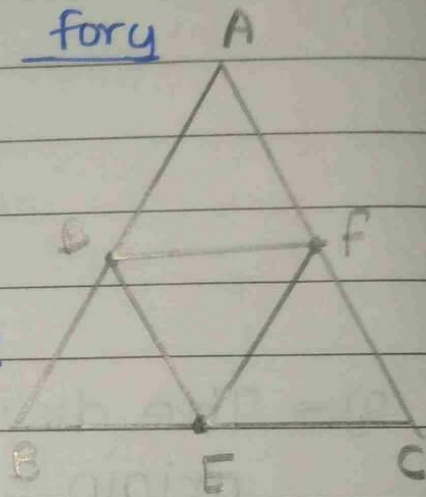


4) $\rightarrow$ Section formula for x and y coordinate when $P(x, y)$ divide line segment AB in ratio $m_1 : m_2$

$$= \left[\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right]$$

for x for y

5) $\rightarrow$ According to question
 D, E and F are the midpoints of AB, BC, AC



By midpoint theorem,

$$FE \parallel AB \text{ and } FE = \frac{1}{2} AB \quad \text{--- ①}$$

So, ~~FE~~ $FE \parallel AB$.

Then, ~~A~~ $FE \parallel BD$

$$BD = \frac{1}{2} AB \quad \text{--- ②}$$

From 1 and 2,

$$FE = BD \text{ and } FE \parallel BD$$

As one pair of sides are parallel and equal. So, BEFD is parallelogram

In $\triangle ABC$ and $\triangle EFD$,

$$\angle ABC = \angle DEF$$

$$\angle ACB = \angle EDF \quad [\text{opposite angles of } \parallel \text{gram}]$$

By AA, $\triangle ABC \sim \triangle EFD$

so, $\frac{\Delta \text{Area of } EFD}{\Delta \text{Area of } \triangle ABC} = \frac{DF^2}{BC^2}$

$$= \frac{DF^2}{2 DF^2} = \frac{DF}{4 DF}$$

$\therefore$ The ratio is 1:4

5) ~~Let C (0, y) divides the AB in ratio t:k~~

6) Here, $(-3, 2) = (x_1, y_1)$
 $(6, 1) = (x_2, y_2)$ B

By using section formula $m:n = 0:y$

$$(0, y) = \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}$$

$$= \frac{6m - 3n}{m+n} = 0$$

$$= 6m - 3n = 0$$

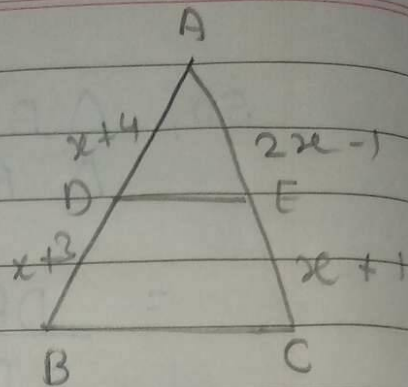
$$= 6m = 3n$$

$$= \frac{m}{n} = \frac{3}{6}$$

$$= \frac{m}{n} = \frac{1}{2}$$

$\therefore$ The ratio is 1:2

7) → Given, $DE \parallel BC$
By using Triangle proportionality theorem,



$$\frac{AD}{BD} = \frac{AE}{EC}$$

$$\frac{x+4}{x+3} = \frac{2x-1}{x+1}$$

By cross multiplication.

$$(x+4)(x+1) = (2x-1)(x+3)$$

$$x^2 + x + 4x + 4 = 2x^2 + 6x - x - 3$$

$$x^2 + 5x - 5x - 7 = 0$$

$$x^2 - 7 = 0$$

$$x^2 = 7$$

$$x = \sqrt{7}$$

8) → Let the required point be $(x, 0)$
since, $(x, 0)$ is equidistant
from the points $(-3, 4)$ & $(2, 5)$

$$= \sqrt{(-3-x)^2 + (4-0)^2}$$

$$= \sqrt{(2-x)^2 + (5-0)^2}$$

$$= \sqrt{9 + x^2 + 16 + 6x}$$

$$= \sqrt{4 + 2x + x^2}$$

$$= \sqrt{4 + x^2 - 4x + 25}$$

$$= x^2 + 6x + 25 = x^2 - 4x + 29$$

$$= 10x = 4$$

$$= x = \frac{4}{10}$$

$$= x = \frac{2}{5}$$

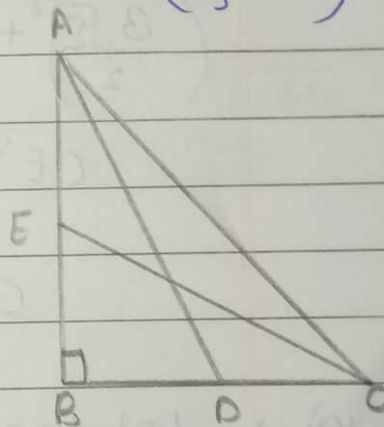
∴ The required point $\left(\frac{2}{5}, 0\right)$

g) → Given,

ABC is right triangle

right angled at B.

Two medians = AD and EC



$$AC = 5 \text{ cm}$$

$$AD = \frac{3\sqrt{5}}{2} \text{ cm}$$

$$AD^2 = AB^2 + BD^2$$

$$AD^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad \left[\because BD = DC \right]$$

$$= AB^2 + \frac{1}{4} BC^2 \quad \text{--- ①}$$

In $\triangle BCE$, right angled at B.

$$CE^2 = BE^2 + BC^2$$

$$CE^2 = BC^2 + \left(\frac{AB}{2}\right)^2$$

$$CE^2 = BC^2 + \frac{1}{4} AB^2 \quad \text{--- ②}$$

Adding ① and ②,

$$AD^2 + CE^2 = AB^2 + \frac{1}{4} BC^2 + BC^2 + \frac{1}{4} AB^2$$

$$AD^2 + CE^2 = \frac{5}{4} (AB^2 + BC^2)$$

$$AD^2 + CE^2 = \frac{5}{4} AC^2$$

Here,

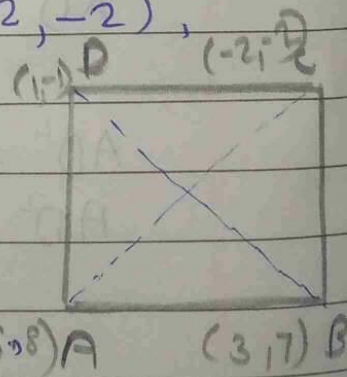
$$\left(\frac{3\sqrt{5}}{2}\right)^2 + CE^2 = \frac{5}{4} \times 25$$

$$CE^2 = \frac{125}{4} - \frac{45}{4} = 20$$

$$CE = \sqrt{20} \text{ cm} = \underline{\underline{2\sqrt{5} \text{ cm}}}$$

10) → Let the points be

A (6, 8) B (3, 7), C (-2, -2),
D (1, -1)



Now,

$$AB = \sqrt{(3-6)^2 + (7-8)^2}$$

$$= \sqrt{(-3)^2 + (-1)^2}$$

$$= \sqrt{9+1}$$

$$= \sqrt{10}$$

$$BC = \sqrt{(-2-3)^2 + (-2-7)^2}$$

$$= \sqrt{(-5)^2 + (-9)^2}$$

$$= \sqrt{25+81}$$

$$= \sqrt{106}$$

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$$\begin{aligned}
 CD &= \sqrt{(1+2)^2 + (-1+2)^2} \\
 &= \sqrt{(3)^2 + (1)^2} \\
 &= \sqrt{9+1} \\
 &= \sqrt{10}
 \end{aligned}$$

$$\begin{aligned}
 AD &= \sqrt{(6-1)^2 + (8+1)^2} \\
 &= \sqrt{(5)^2 + (9)^2} \\
 &= \sqrt{25+81} \\
 &= \sqrt{106}
 \end{aligned}$$

The diagonals.

$$AC = \sqrt{8^2 + 10^2} = \sqrt{64+100} = \sqrt{164}$$

$$BD = \sqrt{2^2 + 8^2} = \sqrt{4+64} = \sqrt{68}$$

Hence, $AB = CD$ and
 $BC = DA$

while $AC \neq BD$.

$\therefore$ It is prove that is a parallelogram